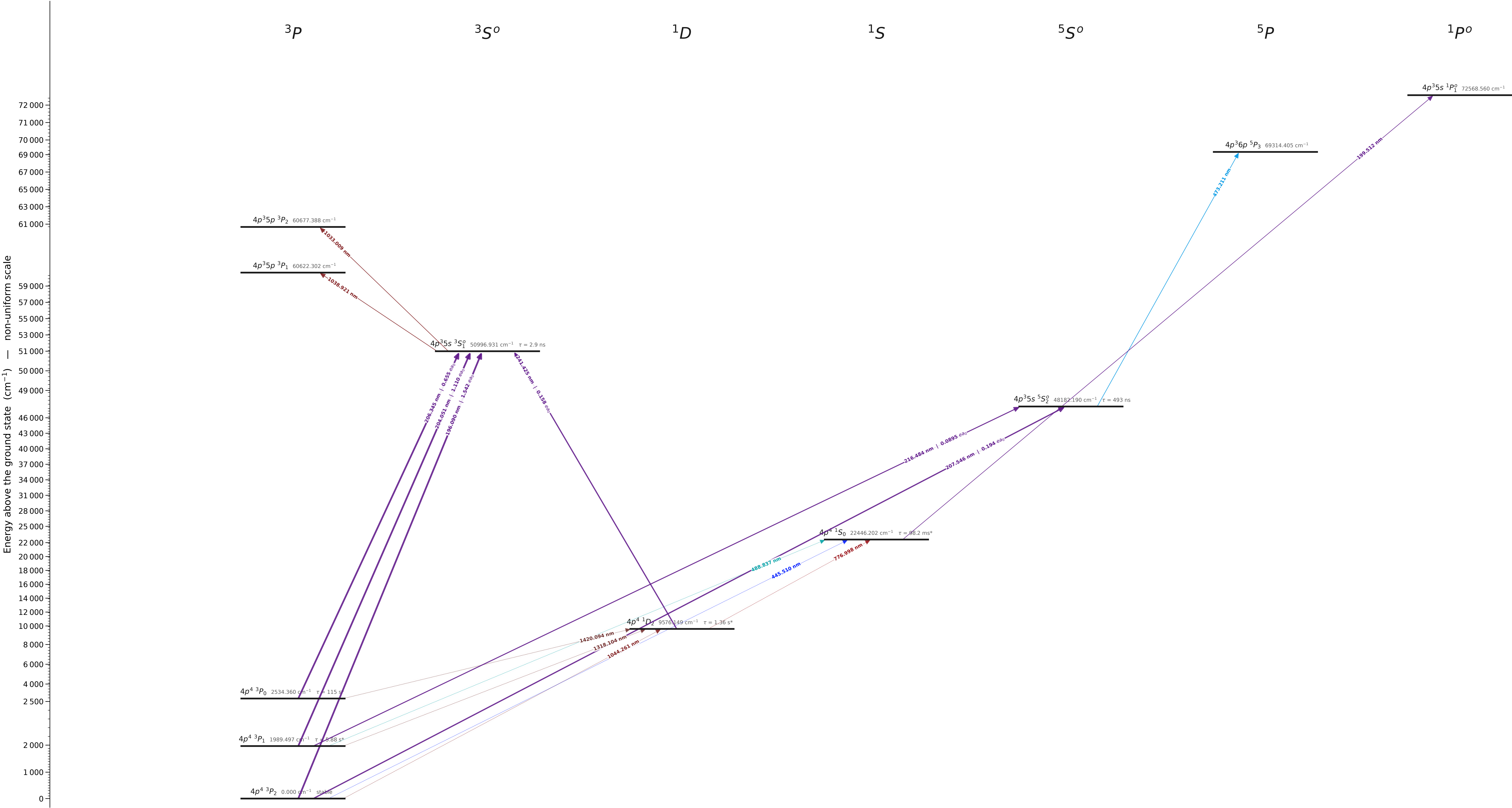
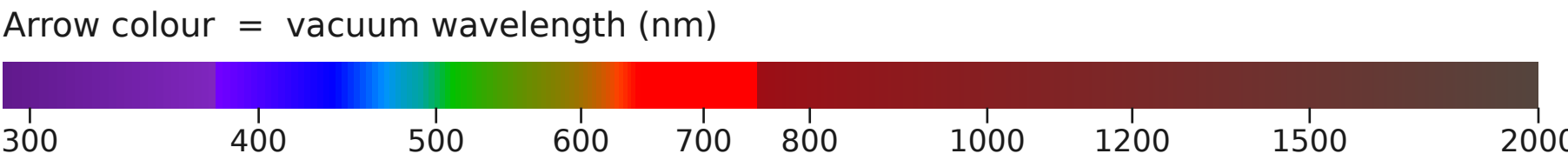


Selenium

Se (Se I). Strongest classified transitions below 2 μm (NIST): 11 levels, 16 lines.

Se⁺ ionisation limit 78658.15 cm⁻¹ = 9.75237 eV (127.132 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| er \| J' \rangle |$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
4p ⁴ 3P ₂ – 4p ³ 5s 3S ₁ ^o	196.0902	–	1528.8495	1.542	2.130e+08	54.88	61.7	resonance line from the ground term
4p ⁴ 1S ₀ – 4p ³ 5s 1P ₁ ^o	199.5118	–	1502.6305	–	–	–	–	classified strong
4p ⁴ 3P ₁ – 4p ³ 5s 3S ₁ ^o	204.0507	203.9852	1469.2059	1.110	9.800e+07	54.88	28.4	resonance line from the ground term
4p ⁴ 3P ₀ – 4p ³ 5s 3S ₁ ^o	206.3448	206.2789	1452.8713	0.655	3.300e+07	54.88	9.52	resonance line from the ground term
4p ⁴ 3P ₂ – 4p ³ 5s 5S ₂ ^o	207.5456	207.4794	1444.4657	0.194	1.700e+06	0.3228	84	207.5 nm intercombination line
4p ⁴ 3P ₁ – 4p ³ 5s 5S ₂ ^o	216.4845	216.4165	1384.8221	0.090	3.200e+05	0.3228	16	216.4 nm intercombination line
4p ⁴ 1D ₂ – 4p ³ 5s 3S ₁ ^o	241.4247	241.3513	1241.7638	0.158	1.200e+06	54.88	0.34	weak branch of 3S°1 to the metastable 1D2 level
4p ⁴ 3P ₂ – 4p ⁴ 1S ₀	445.5097	445.3846	672.9202	–	1.800e-01	1.621e-06	1.77	forbidden line within the ground configuration
4p ³ 5s 3S ₂ ^o – 4p ³ 6p 5P ₃	473.2112	473.0788	633.5279	–	–	–	–	classified strong
4p ⁴ 3P ₁ – 4p ⁴ 1S ₀	488.8373	488.7008	613.2766	–	7.700e+00	1.621e-06	75.6	forbidden line within the ground configuration
4p ⁴ 1D ₂ – 4p ⁴ 1S ₀	776.9976	776.7838	385.8345	–	2.300e+00	1.621e-06	22.6	forbidden line within the ground configuration
4p ³ 5s 3S ₁ ^o – 4p ³ 5p 3P ₂	1033.0091	1032.7260	290.2128	–	–	–	–	classified strong
4p ³ 5s 3S ₁ ^o – 4p ³ 5p 3P ₁	1038.9210	1038.6364	288.5614	–	–	–	–	classified strong
4p ⁴ 3P ₂ – 4p ⁴ 1D ₂	1044.2611	1043.9750	287.0857	–	6.270e-01	1.17e-07	85.3	forbidden line within the ground configuration
4p ⁴ 3P ₁ – 4p ⁴ 1D ₂	1318.1045	1317.7441	227.4421	–	1.100e-01	1.17e-07	15	forbidden line within the ground configuration
4p ⁴ 3P ₀ – 4p ⁴ 1D ₂	1420.0937	1419.7055	211.1075	–	9.200e-05	1.17e-07	0.01	forbidden line within the ground configuration

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).